Ebenezer Gelo

Machine Learning Engineer

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Profile

With a solid foundation in computer science, mathematics, and experience with various machine learning frameworks, I excel in designing algorithms and utilizing deep learning frameworks to solve complex problems. My expertise includes reinforcement learning, natural language processing, and predictive analytics.

Experience

Flyte Agency

Jan 2022 - July 2022 • Co-founder

- Designed and developed a bespoke static website hosted on AWS, providing Flyte Agency with a polished online presence.
- Automated marketing workflows through the implementation of advanced automation tools, boosting efficiency and effectiveness in our campaigns.
- Developed a churn prediction system for a small e-commerce brand, utilizing data analytics and machine learning to optimize customer retention by 40%.

Projects

Learning WVFs without Exploration

Project Link ↗

Compositional RL • Artificial Intelligence

- Researched and implemented offline Reinforcement Learning algorithms, specifically offline DQN and BCQ, to learn goal-oriented value functions (WVF) in varying data buffer sizes.
- Conducted extensive experiments in discrete and continuous action domains using a 2D video game and a robotic environment, evaluating the success rates of learned WVFs with valuable insights into algorithm efficiency under diverse conditions.
- Demonstrated the significance of a large and diverse dataset for learning WVFs in a batch setting, contributing to advancements in offline RL methodologies and their applicability across different domains.

Credit Card Fraud Detection System

Project Link ↗

Deep Learning • Big Data • MLOps

- Developed and implemented a robust Credit Card Fraud Detection System, incorporating advanced machine learning algorithms to proactively identify and prevent fraudulent transactions.
- Integrated MLOps practices into the project, ensuring seamless model deployment, monitoring, and continuous improvement, enhancing the system's accuracy and adaptability over time.

Education

MSc Computer Science

Jan 2024 - Present

University of the Witwatersrand

BSc(Hons) Computer Science

Jan 2023 - Dec 2023

University of the Witwatersrand

BSc Computer Science

Jan 2020 - Dec 2022

University of the Witwatersrand

Skills

- **Languages:** Python, Java, C/C++, Scala, JavaScript
- **ML/DL:** Sci-Kit, TensorFlow, PyTorch
- **Data:** Preprocessing, Feature Engineering, SQL, NoSQL
- **Version Control:** Git, GitLab
- **Cloud:** AWS, Azure, Google Cloud
- **MLOps:** Deployment, Monitoring, Git, CI/CD, Bash
- **Visualization:** Matplotlib, Seaborn, Tableau, Looker, Power BI
- **Stats:** Statistical Analysis

Specialities

- MLOps
- Artificial Intelligence
- Reinforcement Learning
- Large Language Models (LLM)
- Software Engineering

Awards

- Certificate of Class (Wits) x10
- NedBank Educational Bursary